

```
# Upload video
from google.colab import files
uploaded = files.upload()

import cv2
from IPython.display import Video, display

# Get uploaded filename
filename = list(uploaded.keys())[0]

# Open the uploaded video
cap = cv2.VideoCapture(filename)

if not cap.isOpened():
    print("Error: Unable to open the video.")
else:
    # Get video properties
    fps = cap.get(cv2.CAP_PROP_FPS)
    width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
    height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))

    # Create VideoWriters
    fourcc = cv2.VideoWriter_fourcc(*'mp4v')

    slow_out = cv2.VideoWriter(
        "Slow_Motion.mp4",
        fourcc,
        max(fps / 2, 1),      # Half speed
        (width, height)
    )

    fast_out = cv2.VideoWriter(
        "Fast_Motion.mp4",
        fourcc,
        fps * 2,              # Double speed
        (width, height)
    )

    frame_count = 0

    while True:
        ret, frame = cap.read()
        if not ret:
            break

        # Slow Motion (duplicate every frame)
        slow_out.write(frame)
        slow_out.write(frame)
```

```
# Fast Motion (keep alternate frames)
if frame_count % 2 == 0:
    fast_out.write(frame)

    frame_count += 1

cap.release()
slow_out.release()
fast_out.release()

print("Processing Completed Successfully!")

# Display output videos
print("\nSlow Motion Video")
display(Video("Slow_Motion.mp4", embed=True))

print("\nFast Motion Video")
display(Video("Fast_Motion.mp4", embed=True))

# Download output videos
files.download("Slow_Motion.mp4")
files.download("Fast_Motion.mp4")
```

[Choose Files](#) WhatsApp00 PM.mp4

WhatsApp Video 2026-07-20 at 2.35.00 PM.mp4(video/mp4) - 599015 bytes, last modified: 7/20/2026 - 100% done

Saving WhatsApp Video 2026-07-20 at 2.35.00 PM.mp4 to WhatsApp Video 2026-07-20 at 2.35.00 PM (1).mp4

Processing Completed Successfully!

Slow Motion Video

0:00



Fast Motion Video

0:00



